AD 502 – Machine Learning

Pre-Reading Material – Lecture 37

Basic Ensemble Techniques: Bagging & Pasting

Pre-Reading Material – Bagging and Pasting

Estimated Reading Time: 5 Minutes

Introduction

Bagging and Pasting are ensemble learning techniques used to improve the performance and stability of machine learning models. Both techniques train multiple models on different subsets of the same training dataset and combine their predictions to produce a final output.

The main objective of these techniques is to reduce model variance, improve prediction stability, and minimize the risk of overfitting.

Both methods commonly use the same type of machine learning algorithm multiple times, such as Decision Trees.

1. Bagging (Bootstrap Aggregating)

Bagging stands for Bootstrap Aggregating. It is an ensemble learning technique in which multiple training datasets are created by randomly selecting samples from the original dataset with replacement.

With replacement means that the same observation can be selected more than once in a particular subset, while some observations may not be selected.

Working Principle of Bagging:

1. Start with an original training dataset.
2. Create multiple random subsets using sampling with replacement.
3. Train a separate machine learning model on each subset.
4. Generate predictions from all trained models.
5. Combine the predictions to obtain the final output.

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For classification problems, majority voting is generally used, whereas averaging is commonly used for regression problems.

Example:

Suppose a dataset contains information about 100 patients. Multiple bootstrap samples are created from this dataset, and separate Decision Tree models are trained on these samples.

Each model predicts whether a new patient has a particular disease. The final classification is determined through majority voting.

2. Pasting

Pasting is similar to Bagging, but it uses random sampling without replacement.

This means that once an observation is selected for a particular subset, it cannot be selected again within that same subset.

Working Principle of Pasting:

1. Start with the original training dataset.
2. Create multiple random subsets using sampling without replacement.
3. Train separate machine learning models on each subset.
4. Collect predictions from all trained models.
5. Combine the predictions to generate the final output.

Pasting also uses majority voting for classification and averaging for regression.

3. Difference Between Bagging and Pasting

Basis	**Bagging**	**Pasting**
Sampling method	With replacement	Without replacement
Duplicate observations	Possible within a subset	Not possible within a subset
Dataset selection	Bootstrap sampling	Random sampling
Model training	Multiple models	Multiple models
Final prediction	Voting or averaging	Voting or averaging
Main objective	Reduce variance	Reduce variance

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4. Advantages of Bagging and Pasting

1. Improve the stability of machine learning predictions.
2. Reduce model variance.
3. Help control overfitting, particularly for high-variance models.
4. Allow multiple models to learn from different subsets of data.
5. Support parallel model training because individual models can be trained independently.

5. Applications

Bagging and Pasting are useful in several machine learning applications:

- Medical diagnosis and disease prediction.
- Fraud detection in financial transactions.
- Customer behavior classification.
- Credit risk assessment.
- Image classification.
- Predictive analytics.

6. Relationship with Random Forest

Random Forest is a popular ensemble learning algorithm based on the Bagging concept. It trains multiple Decision Trees using bootstrap samples and introduces additional randomness by selecting subsets of features at each split.

This combination helps improve model diversity and prediction performance.

Conclusion

Bagging and Pasting are important ensemble learning techniques that combine multiple models trained on different subsets of a dataset. The major difference between them is their sampling strategy: Bagging uses sampling with replacement, whereas Pasting uses sampling without replacement. Both techniques help improve prediction stability and reduce variance.

Key Terms to Remember: Ensemble Learning, Bagging, Pasting, Bootstrap Sampling, Sampling with Replacement, Sampling without Replacement, Aggregation, Model Variance, Random Forest.

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